

Team Doom and Gloom(Srisailesh, Aditya More , Avaneesh)

Project Report

Github Link: <https://github.com/Hellblazer704/-Hybrid-Neuro-Symbolic-Approach-to-Root-Cause-Identification-in-Customer-Service-Interactions>

EVALUATION METRICS

Overall Accuracy: 99.35%

Macro-Averaged Metrics:

Precision: 97.91%

Recall: 99.86%

F1-Score: 98.87%

Weighted-Averaged Metrics:

Precision: 99.37%

Recall: 99.35%

F1-Score: 99.36%

Per-Class Metrics:

State Name	Precision	Recall	F1	Support
S0 Greeting/Neutral	100.00%	99.18%	99.59%	9879
S1 Info_Exchange	96.84%	100.00%	98.40%	920
S2 Problem_Statement	97.02%	100.00%	98.49%	554
S3 Solution_Offer	96.91%	100.00%	98.43%	878
S4 Friction/Pushback	99.29%	100.00%	99.64%	140
S5 Escalation/Anger	97.40%	100.00%	98.68%	187

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SAMPLE CONVERSATION 1

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DETAILED CONVERSATION ANALYSIS

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Conversation Transcript:

Turn 0: [Agent] - State: S0 (Greeting/Neutral)

"Thank you for calling UnityCredit. This is Brian. How may I assist you today?"

Turn 1: [Customer] - State: S0 (Greeting/Neutral)

"I need to speak with a manager immediately. This is the three time you've sent me a damaged product. This is completely unacceptable."

Turn 2: [Agent] - State: S0 (Greeting/Neutral)

"I'm very sorry to hear about this issue. I'd be happy to help you resolve this right away. Can you tell me more about what happened?"

Turn 3: [Customer] - State: S0 (Greeting/Neutral)

"What happened is that your company keeps sending damaged products. I've been patient, but this is ridiculous. My lawyer has advised me to document everything."

Turn 4: [Agent] - State: S0 (Greeting/Neutral)

"I completely understand your frustration, and I want to make this right. Let me pull up your order history so I can see what's been going on."

Turn 5: [Customer] - State: S3 (Solution_Offer)

"Every time I call, I get promises that it will be fixed, but then the same thing happens again. I'm done being patient."

Turn 6: [Agent] - State: S0 (Greeting/Neutral)

"I can see your account now, and I see you've had three orders with issues. This is definitely not the experience we want you to have. I'm going to personally ensure this gets resolved."

Turn 7: [Customer] - State: S0 (Greeting/Neutral)

"How? You're going to say the same things the other representatives said. I want a manager, and I want replacement for all the time and frustration this has caused."

Turn 8: [Agent] - State: S3 (Solution_Offer)

"I'm going to connect you with my supervisor right now. And before I do, let me document that we'll be providing you with a full replacement and additional compensation for your trouble."

Turn 9: [Customer] - State: S0 (Greeting/Neutral) ⚠️ ROOT CAUSE

"Fine. But if this doesn't get resolved today, I'm moving forward with legal action."

Turn 10: [Agent] - State: S0 (Greeting/Neutral)

"I understand completely. My supervisor is Thomas, and they'll have full authority to resolve this for you. Please hold while I transfer you."

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ROOT CAUSE ANALYSIS

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✓ No escalation detected

Root Cause Identification:

Turn: 9

Speaker: Customer

State: Greeting/Neutral

Risk Jump: 0.015

Risk Before: 0.007

Risk After: 0.021

Critical Turn:

"Fine. But if this doesn't get resolved today, I'm moving forward with legal action."

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SAMPLE CONVERSATION 2

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DETAILED CONVERSATION ANALYSIS

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Conversation Transcript:

Turn 0: [Agent] - State: S0 (Greeting/Neutral)

"Good morning, Wellness Clinic. This is Rebecca. How may I help you?"

Turn 1: [Customer] - State: S0 (Greeting/Neutral)

"Hi, I had blood work done last week and I was told the results would be available in 3-5 days. It's been six days and I haven't heard anything."

Turn 2: [Agent] - State: S1 (Info_Exchange) ⚠️ ROOT CAUSE

"Let me check on those results for you. Can I have your name and date of birth?"

Turn 3: [Customer] - State: S0 (Greeting/Neutral)

"James Miller, born August 3rd, 1972."

Turn 4: [Agent] - State: S0 (Greeting/Neutral)

"Thank you. I'm looking at your chart... I can see the lab results came in yesterday, but they haven't been reviewed by your physician yet."

Turn 5: [Customer] - State: S0 (Greeting/Neutral)

"When will the doctor review them?"

Turn 6: [Agent] - State: S0 (Greeting/Neutral)

"Dr. Chen reviews lab results every afternoon. You should receive a call or portal message by end of day today."

Turn 7: [Customer] - State: S0 (Greeting/Neutral)

"Okay, good. While I have you, I also need a referral to an orthopedic specialist. My knee has been bothering me for months."

Turn 8: [Agent] - State: S0 (Greeting/Neutral)

"I can help with that. Have you discussed this with Dr. Chen already?"

Turn 9: [Customer] - State: S3 (Solution_Offer)

"Yes, at my last appointment two weeks ago. She said she'd send the referral but I haven't received it."

Turn 10: [Agent] - State: S0 (Greeting/Neutral)

"Let me check... I don't see a referral in the system. I'll send a message to Dr. Chen's nurse to process it today."

Turn 11: [Customer] - State: S0 (Greeting/Neutral)

"How will I know which specialist to see?"

Turn 12: [Agent] - State: S3 (Solution_Offer)

"Once the referral is approved, we'll send you a list of in-network orthopedic specialists in your area. You can choose from that list."

Turn 13: [Customer] - State: S0 (Greeting/Neutral)

"Great. One last thing - I got a bill for \$75 from my last visit but I thought my copay was only \$25."

Turn 14: [Agent] - State: S2 (Problem_Statement)

"Let me pull up your billing... I see the charge. The \$25 was your copay, and the additional \$50 was for an additional service performed during that visit."

Turn 15: [Customer] - State: S0 (Greeting/Neutral)

"What service? I just had a routine checkup."

Turn 16: [Agent] - State: S0 (Greeting/Neutral)

"It shows here that Dr. Chen performed a skin lesion removal. Do you remember that?"

Turn 17: [Customer] - State: S0 (Greeting/Neutral)

"Oh yes, she removed a small mole. I didn't realize that would be extra."

Turn 18: [Agent] - State: S2 (Problem_Statement)

"Minor procedures like that are billed separately from the office visit. The \$50 is actually quite reasonable for that service."

Turn 19: [Customer] - State: S0 (Greeting/Neutral)

"Okay, that makes sense now. Will my insurance cover any of it?"

Turn 20: [Agent] - State: S0 (Greeting/Neutral)

"Your insurance already processed their portion. The \$50 is your responsibility after insurance."

Turn 21: [Customer] - State: S0 (Greeting/Neutral)

"Alright. So my test results should be available today, the referral will be processed, and the \$75 bill is correct?"

Turn 22: [Agent] - State: S1 (Info_Exchange)

"Exactly. You'll get a call about your results, the referral information will be sent to your email, and the billing is accurate."

Turn 23: [Customer] - State: S0 (Greeting/Neutral)

"Perfect. Thank you for clarifying everything."

Turn 24: [Agent] - State: S0 (Greeting/Neutral)

"You're very welcome. Have a great day!"

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ROOT CAUSE ANALYSIS

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✓ No escalation detected

Root Cause Identification:

Turn: 2

Speaker: Agent

State: Info_Exchange

Risk Jump: 0.006

Risk Before: 0.000

Risk After: 0.006

Critical Turn:

"Let me check on those results for you. Can I have your name and date of birth?"

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SAMPLE CONVERSATION 3

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DETAILED CONVERSATION ANALYSIS

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Conversation Transcript:

Turn 0: [Agent] - State: S0 (Greeting/Neutral)

"Thank you for calling GuardianPolicy. This is Emma. How may I help you?"

Turn 1: [Customer] - State: S0 (Greeting/Neutral)

"I received a denial letter for a medical claim and I'm really confused because I know this procedure is covered under my plan. It was for a prescription medication for my knee."

Turn 2: [Agent] - State: S0 (Greeting/Neutral)

"I'm sorry to hear about the denial. I'll be happy to review this with you. Can I get your member ID number?"

Turn 3: [Customer] - State: S0 (Greeting/Neutral)

"It's 36175204. The claim number is 246233."

Turn 4: [Agent] - State: S0 (Greeting/Neutral) ⚠️ ROOT CAUSE

"Thank you. Let me pull up your claim... I see the claim for the prescription medication. According to the notes, it was denied due to lack of formulary status."

Turn 5: [Customer] - State: S0 (Greeting/Neutral)

"But my doctor said everything was submitted correctly. They've done this many times before with your company."

Turn 6: [Agent] - State: S0 (Greeting/Neutral)

"I understand your frustration. Let me check the details... I'm looking at what was submitted. It appears the formulary status documentation was submitted, but it was submitted three days after the procedure was performed."

Turn 7: [Customer] - State: S0 (Greeting/Neutral)

"So it was denied just because of timing? That seems really unfair. I was in pain and needed the prescription medication right away."

Turn 8: [Agent] - State: S0 (Greeting/Neutral)

"I completely understand. In cases where there's a medical emergency or urgent medical necessity, we can file an appeal. Did your doctor document that this was urgent or causing significant pain?"

Turn 9: [Customer] - State: S0 (Greeting/Neutral)

"Yes, absolutely. I had been dealing with severe knee pain for weeks, and it was affecting my ability to work."

Turn 10: [Agent] - State: S0 (Greeting/Neutral)

"That's important information. What I can do is initiate a formulary status appeal for you right now. I'll need to get some documentation from your provider's office, but this type of appeal usually has a good chance of approval."

Turn 11: [Customer] - State: S0 (Greeting/Neutral)

"How long will that take?"

Turn 12: [Agent] - State: S0 (Greeting/Neutral)

"The appeal process typically takes 30 business days, but I can request an expedited review which would be 72 hours. In the meantime, please don't pay the bill - wait for the appeal decision."

Turn 13: [Customer] - State: S0 (Greeting/Neutral)

"Okay, thank you. I was worried I'd be stuck with a huge bill."

Turn 14: [Agent] - State: S1 (Info_Exchange)

"You're welcome. I'm sending the appeal request now, and you'll receive a confirmation email. I'll also note that you were experiencing significant pain requiring urgent pharmaceutical coverage. Is there anything else I can help clarify?"

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ROOT CAUSE ANALYSIS

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✓ No escalation detected

Root Cause Identification:

Turn: 4

Speaker: Agent

State: Greeting/Neutral

Risk Jump: 0.000

Risk Before: 0.006

Risk After: 0.006

Critical Turn:

"Thank you. Let me pull up your claim... I see the claim for the prescription medication. According to the notes, it was denied due to lack of formulary status."

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SAMPLE CONVERSATION 4

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DETAILED CONVERSATION ANALYSIS

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Conversation Transcript:

Turn 0: [Agent] - State: S0 (Greeting/Neutral)

"ConnectNet retention department, this is Marcus. How can I help you today?"

Turn 1: [Customer] - State: S2 (Problem_Statement)

"I've had enough. My internet has been down three times this month and I'm being overcharged. I want to cancel immediately."

Turn 2: [Agent] - State: S0 (Greeting/Neutral)

"I'm very sorry to hear about your experience. Let me pull up your account to see what's been happening."

Turn 3: [Customer] - State: S3 (Solution_Offer)

"You know what's been happening. I've called four times already. Each time I'm told it's fixed, but it's not."

Turn 4: [Agent] - State: S0 (Greeting/Neutral)

"I can see your service history, and you're absolutely right. You've had three outages in 30 days and there are billing discrepancies."

Turn 5: [Customer] - State: S0 (Greeting/Neutral)

"So what are you going to do? Because FiberNet is offering me a better deal and their service is reliable."

Turn 6: [Agent] - State: S0 (Greeting/Neutral)

"I completely understand. Before you make that decision, let me see what I can offer to keep your business."

Turn 7: [Customer] - State: S0 (Greeting/Neutral)

"I've heard promises before. Why should I believe you this time?"

Turn 8: [Agent] - State: S3 (Solution_Offer)

"I'm going to take action right now. First, I'm crediting three months of service to your account - that's \$225."

Turn 9: [Customer] - State: S0 (Greeting/Neutral)

"That's something, but what about the reliability issues? Money doesn't help when I can't work from home."

Turn 10: [Agent] - State: S0 (Greeting/Neutral)

"You're absolutely right. I'm also upgrading you to our business-class service at no additional cost. It has 99.9% uptime guarantee."

Turn 11: [Customer] - State: S2 (Problem_Statement)

"What happens if it goes down again?"

Turn 12: [Agent] - State: S2 (Problem_Statement) ⚠️ ROOT CAUSE

"With business-class service, we provide 4G backup internet automatically. If your main line fails, you'll still have connectivity."

Turn 13: [Customer] - State: S0 (Greeting/Neutral)

"That actually sounds useful. But I'm still frustrated about all the time I've wasted dealing with this."

Turn 14: [Agent] - State: S0 (Greeting/Neutral)

"I understand completely. I'm also adding six months of our premium support package, which gives you priority response within 2 hours."

Turn 15: [Customer] - State: S0 (Greeting/Neutral)

"Okay, that's better than what FiberNet offered. But if there's one more outage, I'm gone."

Turn 16: [Agent] - State: S2 (Problem_Statement)

"That's fair. I'm making a note in your account that any future issues should be escalated to me personally. Here's my direct extension."

Turn 17: [Customer] - State: S0 (Greeting/Neutral)

"Alright, I'll give you one more chance. But I'm holding you to this."

Turn 18: [Agent] - State: S1 (Info_Exchange)

"I appreciate that. Your upgrade will be active within 24 hours and you'll receive confirmation via email."

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ROOT CAUSE ANALYSIS

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✓ No escalation detected

Root Cause Identification:

Turn: 12

Speaker: Agent

State: Problem_Statement

Risk Jump: 0.006

Risk Before: 0.000

Risk After: 0.006

Critical Turn:

"With business-class service, we provide 4G backup internet automatically. If your main line fails, you'll still have connectivity."

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SAMPLE CONVERSATION 5

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DETAILED CONVERSATION ANALYSIS

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Conversation Transcript:

Turn 0: [Agent] - State: S0 (Greeting/Neutral)

"Thank you for calling DeviceCare technical support. This is Joshua. How can I assist you?"

Turn 1: [Customer] - State: S0 (Greeting/Neutral)

"Hi, I updated my design software yesterday and now it freezes during startup I try to launch it."

Turn 2: [Agent] - State: S0 (Greeting/Neutral)

"I'm sorry to hear you're experiencing issues after the update. That's definitely frustrating. Can you tell me which version you updated to?"

Turn 3: [Customer] - State: S0 (Greeting/Neutral)

"I think it was version 12.3. The update installed automatically."

Turn 4: [Agent] - State: S0 (Greeting/Neutral)

"I see. There have been some reported compatibility issues with that version on certain system configurations. Let me check your system setup. Are you on Windows or Mac?"

Turn 5: [Customer] - State: S0 (Greeting/Neutral)

"macOS Monterey. I have a fairly new computer, so I don't think that's the issue."

Turn 6: [Agent] - State: S0 (Greeting/Neutral)

"Let me look into this... The issue seems to be related to the rendering engine in the new version conflicting with your graphics drivers. This is affecting a small percentage of users."

Turn 7: [Customer] - State: S0 (Greeting/Neutral)

"Okay, so how do I fix it? I need this software for work and I have deadlines."

Turn 8: [Agent] - State: S3 (Solution_Offer)

"I completely understand the urgency. I can guide you through rolling back to the previous version, which was stable. Alternatively, there's a hotfix patch that was just released this morning that should resolve the issue. Which would you prefer?"

Turn 9: [Customer] - State: S2 (Problem_Statement)

"If there's a fix available, I'd rather try that first instead of downgrading."

Turn 10: [Agent] - State: S3 (Solution_Offer)

"Absolutely. I'm sending you a link to download the hotfix patch right now. It's a small file, about 127MB. Once you download it, run the installer and it will patch the rendering engine issue."

Turn 11: [Customer] - State: S0 (Greeting/Neutral)

"Okay, I'm downloading it now. Should I uninstall the current version first?"

Turn 12: [Agent] - State: S0 (Greeting/Neutral)

"No, don't uninstall. The patch will install over the existing version. Just make sure the design software is closed when you run the patch installer."

Turn 13: [Customer] - State: S2 (Problem_Statement) ⚠️ ROOT CAUSE

"Got it. The download is complete. I'm running it now."

Turn 14: [Agent] - State: S0 (Greeting/Neutral)

"Perfect. The installation should take about 2-3 minutes. Once it's done, try launching the design software again. I'll stay on the line with you."

Turn 15: [Customer] - State: S0 (Greeting/Neutral)

"Okay, the patch is installed... Let me try opening the software... It's working! Thank you so much!"

Turn 16: [Agent] - State: S0 (Greeting/Neutral)

"Excellent! I'm glad we got that resolved. If you experience any other issues, don't hesitate to call back. Is there anything else I can help with today?"

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ROOT CAUSE ANALYSIS

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✓ No escalation detected

Root Cause Identification:

Turn: 13

Speaker: Customer

State: Problem_Statement

Risk Jump: 0.006

Risk Before: 0.000

Risk After: 0.006

Critical Turn:

"Got it. The download is complete. I'm running it now."

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PIPELINE COMPLETE

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Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.

 Scanning for available Gemini models...

 Connected to Gemini Model: models/gemini-2.5-flash

 Analyzing Conversation...

 Call Analyzed. Root Cause Identified.

 Session Started. Type 'exit' to quit or 'report' to save.

Analyst: what is the cause

AI: The agent's unhelpful and dismissive response "I can't fix it right now. You have to wait." in Turn 3.

Analyst: what is ur suggestion to improve

AI: The agent should acknowledge the customer's frustration, explain **why** they can't fix it immediately (e.g., "there's a known outage"), and provide clear next steps or an estimated resolution timeframe instead of a dismissive "You have to wait."

Analyst: is there any feed back you would like to provide

AI: The agent's dismissive response in Turn 3 ("I can't fix it right now. You have to wait.") directly escalated the situation. The agent should have acknowledged the customer's frustration, explained the reason for the delay, and provided a clear next step or estimated resolution time instead of a blunt dismissal.

Analyst: report

 Report saved to Session_Report.txt

Analyst: exit